

Impact Statement

Localized Agronomic Knowledge Retrieval in Nigeria

This project builds a retrieval system for a Retrieval-Augmented Generation pipeline that matches a Nigerian smallholder farmer's plain language question, such as a description of a discolored leaf or a wilting stem, to the most relevant document in a curated agricultural extension knowledge base. It addresses a specific gap: agronomic knowledge already exists in extension documents and research publications, but farmers cannot search it, because keyword search does not bridge the way farmers describe problems and the way experts document solutions.

A working retriever lowers the barrier between a farmer's question and expert guidance, reducing dependence on the availability of an extension officer. Because the system is designed to reward intent-correct retrieval rather than surface keyword overlap, it can distinguish a document about preventing a disease from one about treating it, which matters when a farmer needs to act on the right advice quickly. Extension officers benefit too, since a reliable first pass at document search frees their time for direct farmer support rather than manual searching through reference material.

A high ranking score on a system like this reflects retrieval quality against benchmark labels, not the correctness of the underlying agronomic content, so a well-performing system could still surface confidently wrong information if deployed without review. Misretrieval carries real economic weight here: returning a generic soil-health document for an urgent disease query could contribute to crop or income loss. There is also a language and access gap, since the system and its documents are English-only, while many of the farmers it is meant to serve are more comfortable in Hausa, Yoruba, Igbo, or Pidgin. The knowledge base may also not represent every regional crop, pest, or local term across Nigeria's agro-ecological zones, and a well-scoring retriever cannot tell the difference between a document that was missed and one that simply does not exist in the corpus.

The project's design choices are meant to protect farmer autonomy and extension officer dignity. The goal is to support, not replace, an extension officer's judgment, and to give farmers information they can act on rather than a single forced answer. Fairness matters directly here, since retrieval quality should not degrade because a farmer uses dialect or non-standard phrasing, and community wellbeing depends on the system being trustworthy enough that neither farmers nor officers feel misled by it.

Planned mitigations include a staged rollout that starts with extension officers reviewing retrieved documents before any direct farmer-facing deployment for high-stakes categories, a separate content-accuracy review layered on top of retrieval-quality evaluation, and evaluation broken down by region and query type rather than a single aggregate score, so uneven performance for particular dialects or crops stays visible instead of hidden behind a good average.

This assessment rests on the project's design intentions and the reference dataset's own documentation, not an independent audit of retrieval accuracy in the field. How well the system performs for underrepresented dialects or locally specific crop and pest names remains untested until real farmer

queries are collected and reviewed.

Word count: approximately 500 words.